

>help

• 37 commands – the one line here it knows

- [1] aaron s, blackwell af 2013. from sonic pi to overtone. farm'13.
- [2] berthaut f et al 2013. rouages: revealing the mechanisms of digital musical instruments to the audience. nine'13.
- [3] burland k, mclean a 2016. understanding live coding events. ijpdm 12(2).
- [4] collins n, mclean a, rohrlhuber j, ward a 2003. live coding in laptop performance. organised sound 8(3).
- [5] collins n, mclean a 2014. algorave. nine'14.
- [6] eno b 1996. generative music 1. sseyo koan.
- [7] gibson w 1984. neuronancer. ace.
- [8] jack o 2017. hydra. software; iclc'19 demo.
- [9] lovelace aa 1843. notes to menabrea. scientific memoirs 3.
- [10] matheus mv 1963. the digital computer as a musical instrument. science 142.
- [11] mccartney j 2002. rethinking the computer music language: supercollider. onj 26(4).
- [12] mclean a 2014. making programming languages to dance to: live coding with tidal. farm'14.
- [13] moon t 1977. sideburns 1, p2.
- [14] obarski k 1987. ultimate soundtracker; goriunova o 2012, p102.
- [15] reeves s, benford s, o'malley o, fraser n 2005. designing the spectator experience. chi'05.
- [16] reich s 1968. music as a gradual process.
- [17] riley t 1964. in o. sf tape music center.
- [18] roos f, mclean a 2023. strudel: live coding patterns on the web. iclc'23.
- [19] schloss wa 2003. using contemporary technology in live performance: the dilemma of the performer. jnmr 32(3).
- [20] spiegel l 1986. music mouse.
- [21] toplap 2004. manifestodraft; ward a et al, read_me.
- [22] toussaint g 2005. the euclidean algorithm generates traditional musical rhythms. bridges.
- [23] weingart w 2000. typography. lars muller.
- [24] xenakis i 1971. formalized music. indiana up.

set in the deck's own 12x24 and 6x12 faces; two inks, one bit each; every picture drawn by its engine. a deck after gibson [7], a grid after weingart, sometimes called swiss punk [23]. full references: zine/README.md. copy it, fold it, pass it on.

.....X

cYbErDeCk
zine #0

>hello

a getting-started zine
for a box
that plays text.

X.....

the engine
"might compose elaborate
and scientific pieces of
music of any degree of
complexity or extent."

ada lovelace, 1843 [9]

02

•X••••••••••

>room

• room? try: help

1843	lovelace	an engine could compose [9]
1963	mathews	the computer, an instrument [10]
1964	riley	in c: 53 phrases, repeated [17]
1968	reich	music as a gradual process [16]
1971	xenakis	formalized music [24]
1986	spiegel	music mouse [20]
1987	obarski	the tracker [14]
1996	eno	generative music 1 [6]
2002	mccartney	supercollider 3 [11]
2003	collins et al	live coding [4]
2004	toplap	show us your screens [21]
2005	toussaint	euclid: open here still [22]
2013	aaron, blackwell	sonic pi [1]
2014	mclean	tidal [12]
2014	collins, mclean	algorave [5]
2017	jack	hydra [8]
2023	roos, mclean	strudel [18]
2026	you	this box

••••••••••X•

15

>what

• what? try: help

a keyboard, a screen that
needs daylight, a clock.

for anyone who lost a
night to a tracker, a
patch cable or a
semicolon.

x is a hit. . is a rest.
now form a band [13].

on the deck: one set in three acts - ground, lift,
orbits; run them in order; the set ends in silence.

.....X..

..X.....

03

you type lines.

a line is a lane.

lanes play: drums, synths,
filters, pictures.

no mouse. no menu. no mode.

the page you are editing
is the page that plays.

>pictures

• pictures? try: help

six fields, ten operators,
no shapes.

fields: disco box turn ramp grid noise
levels: mask edge
bends: echo move spin warp grow thin flip fold

a ring is a disc through an
edge: >disc 8 >edge 1



orbitals, drawn by the deck: a planet, x at 16 steps and
y at 12, round a sun on the beat; night, day, eclipse.
echo keeps the trail: feedback is the trick [8].

04

...X.....

13

.....X...

a lane that ends can start
another. no timeline: an
arrangement is endings.

>knob1 = knob and >osc in 9000: a phone, a laptop or
another deck sends /deck/knob1 and >route out knob1
follows it. the value lives here, so losing the phone
loses nothing.

the deck talks back:
♦ _ needs a note before it
♦ a chord goes in []: [0,4,7]
♦ a name is letters: conga
♦ 16 lanes is all there is.
♦ run cannot run itself

laptop sets taught crowds
to read projected code as
proof that it was live
[21][4][3].

here the gesture and the
sound are one thing to
watch [19][15][2].

an esp32-s3 and a 400x300 reflective panel, one bit, no
backlight. 96 ticks a beat, 16 lanes, 37 verbs. usb midi
jitter at the host: 0.03 ms, measured. it makes no sound
itself: it sends midi, oso and pictures to what does.

12

.....X.....

05

....X.....

>hello

• hello? try: help

>kick 9...8...9...8...



velocity is ink

>play

>names

• names? try: help

the names are yours:

>conga = note 63

>strings = voice 3 ch 5

>fx = cc 20 ch 2

>strings = ch 6 moved.

>conga = gone.

anything drives anything:

>route cut kick

>route disc kick

>tom x.x.x.x. !2

>route crash tom:end

06

.....X.....

11

.....X.....

```
strudel: bd ~ sd ~  
here:    x.x.
```

51 of 51 corpus patterns play what strudel 1.2.6 plays,
note for note [18][12].

what the deck will not play, it refuses, and boxes the
character:

```
>hat x...X...  X is gone: 9 is loud  
>hat x...?...  ? is gone: x%50 is maybe  
>hat x.-...    - is not a rest here: .
```

```
ctrl+enter  runs the line  
enter       makes a line
```

again, while it plays:
gone. again: back.

>toggle kick hat
a block, off and back.

it sends midi, not sound.

```
>usb on  to a computer:  
         the deck restarts  
         as a midi device.  
>din 17  to a synth, with  
         no computer. build  
         the cable first.
```

drums on channel 10, bass 1, lead 2, pad 3, arp 4: the
names are lines in the boot document. change them there.

10

.....X.....

07

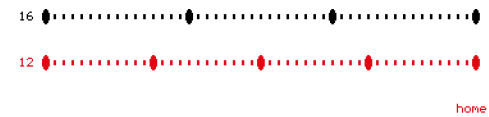
.....X.....

>grammar

• grammar? try: help

x a hit
.
0-9 how much
_ hold the last one
[xx] two in one step
[0,4,7] all at once
<a b> one each time round
<00 77> a word each time
x%30 30 times in 100
/2 *2 this lane's speed
!4 four times, then end

a line is as long as it is.



16 against 12 comes home
every 3 bars. that is all
polyrhythm is.

08

.....X.....

09

.....X.....